

WELCOME ISLAND, ON, Canada

WMO: 717510

Lat: 48.369N

Lon: 89.119W

Elev: 211

StdP: 98.81

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-24.1	-31.1	0.2	-26.2	-28.4	0.3	-23.6	16.2	-14.1	14.9	-14.2	7.1	300	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	24.1	18.2	22.5	17.5	21.1	17.0	19.7	22.1	18.6	21.1	17.7	20.1	5.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	21.0	17.6	13.0	19.8	16.7	12.2	19.1	57.3	22.1	53.6	21.2	50.9	20.1	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	13.0	11.5	-28.7	27.2	3.1	1.9	-30.9	28.6	-32.7	29.7	-34.4	30.8	-36.7	32.2
			-29.2	21.5	3.1	2.0	-31.4	22.9	-33.2	24.0	-35.0	25.1	-37.2	26.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-12.2	-11.0	-4.7	2.1	7.7	13.4	17.0	17.2	13.5	7.2	-0.1	-7.0
	DBStd	11.21	7.10	6.29	5.84	3.74	3.03	2.71	2.55	2.52	3.26	3.81	5.28	6.45
	HDD10.0	2996	688	588	457	236	84	4	0	0	8	103	303	526
	HDD18.3	5382	946	822	715	486	329	149	57	52	146	344	553	784
	CDD10.0	694	0	0	0	1	13	107	218	224	114	18	0	0
	CDD18.3	38	0	0	0	0	0	2	17	18	2	0	0	0
	CDH23.3	82	0	0	0	0	0	8	47	24	3	0	0	0
	CDH26.7	6	0	0	0	0	0	1	4	1	0	0	0	0
Wind	WSAvg	5.5	6.4	5.8	5.4	5.6	5.0	4.4	4.5	4.6	5.0	6.1	6.5	6.7
Precipitation	PrecAvg	708	39	25	34	59	73	83	87	68	72	72	56	41
	PrecMax	940	76	63	58	200	184	185	212	131	129	152	151	87
	PrecMin	523	11	8	14	7	38	18	31	22	25	25	12	17
	PrecStd	97	15	13	13	36	34	41	40	27	28	32	33	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	3.8	8.6	13.2	19.7	24.5	27.0	25.8	23.1	17.9	11.8	5.1
		MCWB	1.6	2.3	5.3	7.9	12.6	17.4	19.1	19.1	16.9	12.9	9.0	3.9
	2%	DB	1.5	1.7	5.2	10.2	16.2	21.4	24.5	23.7	20.5	15.5	9.0	3.9
		MCWB	0.7	0.7	2.7	5.6	10.8	15.7	18.2	18.5	16.6	12.6	7.2	3.0
	5%	DB	0.2	0.1	3.4	8.4	14.0	19.2	22.7	22.3	19.1	14.0	7.9	2.7
		MCWB	-0.5	-0.8	1.5	4.9	9.8	14.7	17.6	17.7	16.3	11.6	6.4	2.0
	10%	DB	-1.6	-1.8	2.2	6.9	12.3	17.7	21.2	21.0	17.9	12.6	6.7	1.5
		MCWB	-2.5	-2.7	0.7	3.8	9.1	13.9	17.2	17.3	15.6	10.5	5.3	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	2.5	5.5	9.0	13.7	18.9	21.3	21.6	19.0	14.7	9.6	4.3
		MCDB	2.7	3.4	8.3	12.3	16.8	22.3	23.9	24.0	20.7	16.0	11.3	4.8
	2%	WB	0.7	0.9	3.0	6.6	12.0	16.7	19.5	20.0	17.7	13.1	7.8	3.1
		MCDB	1.4	1.6	4.6	9.1	14.5	19.6	22.4	22.1	19.3	14.9	8.8	3.7
	5%	WB	-0.5	-0.7	1.9	5.3	10.7	15.5	18.5	18.9	16.8	11.9	6.5	2.1
		MCDB	0.2	0.0	3.1	7.9	13.4	18.6	21.5	21.0	18.3	13.5	7.6	2.6
	10%	WB	-2.5	-2.7	0.9	4.1	9.4	14.5	17.6	17.9	15.9	10.8	5.5	0.8
		MCDB	-1.7	-1.9	2.1	6.5	11.9	17.0	20.6	20.0	17.4	12.4	6.6	1.5
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.7	8.2	6.3	6.6	6.9	6.8	6.1	5.5	4.7	5.0	6.0
		MCDBR	6.3	7.9	7.3	8.3	9.3	9.4	8.9	7.6	6.7	6.4	5.1	5.2
		MCWBR	6.0	7.2	5.6	5.1	5.4	5.4	4.4	4.1	4.2	4.5	4.3	5.0
	5% WB	MCDBR	6.6	7.6	6.9	8.0	8.5	8.7	8.2	6.5	5.7	5.9	4.9	5.3
		MCWBR	6.3	7.1	5.4	5.4	5.7	5.8	4.9	4.1	4.2	4.7	4.3	5.2
Clear-Sky Solar Irradiance	taub	0.264	0.272	0.304	0.343	0.361	0.380	0.397	0.377	0.356	0.316	0.290	0.265	
	taud	2.415	2.389	2.387	2.375	2.410	2.383	2.383	2.438	2.483	2.561	2.530	2.462	
	Ebn at Noon	823	898	916	908	901	882	859	864	850	839	789	777	
	Edh at Noon	64	83	99	111	112	116	114	104	90	71	59	55	
All-Sky Solar Radiation	RadAvg	1.08	2.10	3.39	4.72	5.47	5.88	5.93	5.10	3.66	2.16	1.14	0.88	
	RadStd	0.13	0.22	0.29	0.48	0.36	0.41	0.33	0.38	0.26	0.27	0.11	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page